

CAST: Supplementary Experimental Protocols and Results

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This supplement accompanies the Experiments section. It preserves the complete supporting tables and separates reported measurements from protocol details and unresolved run associations. Results from the main comparisons, component study, alternative updates, and parameter sweeps retain their own reference runs; their different CAST values are not pooled.

S1 Simulation Protocols

Matched DICE-RL and CAST comparisons start from the same pretrained checkpoint and retain the residual architecture, critic-training procedure, replay recipe, and stated interaction budget. DICE-RL’s actor maximizes critic value with selectively applied BC regularization. CAST replaces that actor objective with action-target regression. The comparison is between complete actor updates, including their weighting of replay states. In online refinement, resulting trajectories and learned critic weights need not remain identical.

Replication and evaluation. Robomimic evaluations use 300 episodes. Drifting can and square results average the last three checkpoints separately for training seeds 42–44; the main table reports the range of these seed-level averages. Two DICE-RL can runs end before the common 20K-update budget. LIBERO main results use 100 rollouts for each of evaluation seeds 10000–10002. These quantify evaluation variation and are not three independently trained policies. Both the drifting and flow-matching multitask bases are first trained on all 90 LIBERO tasks. Refinement then targets the same eight challenging tasks for both bases. Each selected task has its own residual and critic, deployed through task-ID routing over the shared frozen multitask base. Reported means cover this selected subset. The matched LIBERO recipe uses value-guided selection among ten candidates. DMC returns average ten episodes of 1000 steps each. Swing-up averages two training seeds; the remaining reported DMC rows each use one.

Actor settings. The shared LIBERO recipe uses $K = 16$ candidates, weak restoration $\eta_0 = 0.05$, radial restoration $\eta_{\text{anc}} = 1$, an RMS radius $\rho = 0.05$, normalized critic-field cap $\delta_Q = 1$, and combined step cap $\delta_{\text{tot}} = 0.15$. The two main-table settings use $\eta_Q = 1$ and 2 across all eight tasks, respectively. This parameter scales actor guidance; it is not the critic optimizer learning rate. Robomimic and default DMC gradient recipes use $\eta_Q = 0.5$ and $\rho = 0$. The DMC quadruped result is an explicitly tuned exception.

S2 Physical Refinement Protocol

Platform and policy. The OpenArm setup operates the seven-joint left follower arm and its parallel gripper. Chest RealSense D435 and wrist D405 cameras provide RGB observations, resized to 128×128 and encoded by a shared ResNet-18. Follower joint/gripper state and a task-language embedding condition a flow-matching policy with a diffusion-transformer backbone. Ten Euler steps generate 30-step chunks of absolute joint/gripper targets. Commands are issued at 15 Hz; a new chunk is requested every eight control steps. The common execution interface limits arm-command changes to 0.6 rad/s and ends attempts after 20 s.

Task-specific data. Policy pretraining uses 40 placement demonstrations and approximately 60 stacking demonstrations. BC refinement uses 30 additional expert demonstrations for each task, separate from the pretraining data. For RL refinement, we collect 40 physical robot rollouts per task and retain all outcomes, including both successes and failures. DICE-RL and CAST share the same task-specific rollout pool and outcome labels. Refinement is performed offline after data collection. These counts are training episodes; the 20 autonomous evaluation attempts per method and task are separate.

Training. BC fine-tunes the base and encoder for 40 additional epochs; the documented peak learning rates are 3×10^{-5} for placement and 10^{-5} for stacking. Both RL arms freeze the base and encoder and train a residual and five-critic ensemble for 6000 critic updates, with 500 critic-only warm-up steps and an actor update every two steps.

They share the data, initialization seed, replay-index schedule, and optimizer budget. The batch size is 32, the actor uses 16 noise samples, and critic hidden layers have width 256. Actor and critic learning rates are 10^{-4} , the target-critic Polyak coefficient is 0.01, the discount is 0.99, and the CQL-style penalty weight is 0.1. DICE-RL uses the residual/Q-filtered BC objective with BC weight 100. CAST uses gradient guidance with $\eta_Q = 1$, $\eta_0 = 0.05$, $\eta_{\text{anc}} = 1$, $\rho = 0.05$ RMS, $\delta_Q = 1$, and $\delta_{\text{tot}} = 0.15$. These are offline physical-refinement settings, not a reproduction of every published online DICE-RL setting. BC versus RL changes data, supervision, and trainable parameters; it is not a matched collection-effort or actor-only comparison.

Replay and evaluation. Saved follower state and issued commands provide the state/action pairs for autonomous attempts. The critic and residual correction operate on the nominal eight-action executed prefix of each 30-action chunk. Behavior latents were not recorded, so the critic is noise-independent; new actor candidates still use same-noise base/residual pairing. Only successful terminal transitions receive reward 1, all other rewards are zero, and nonterminal critic targets bootstrap future value estimates. The autonomous success counts are separate from training losses and held-out imitation errors. There are 20 attempts per task and method; placement poses are randomized on the stairs. Paired resets and a common post-release observation interval are not documented. Historical camera recordings lack per-frame timestamps, so offline image/state alignment is approximate. The reported evaluation counts do not yet identify the exact evaluated checkpoint paths or hashes. Matching the training protocol therefore does not by itself verify the evaluated checkpoints.

S3 Complete Robomimic Results

Table S1: Robomimic success (%). Top: DICE-RL and CAST refine the same pretrained base within each row. Drifting entries span three training seeds, each averaged over its last three 300-episode evaluations; † marks two DICE-RL can runs ending before 20K updates. Diffusion results use 496K environment steps; z denotes value-only CAST with an unspecified interaction budget. Bottom: external square results use separate bases, budgets, and evaluation protocols. Δ is the gain over each method’s own base in percentage points (pp); * marks a stricter success rule and $\circ$ an ongoing run. Bold marks the highest refinement result per row in the top block, and the highest fine-tuned result and gain within each external group.

Task / base class	Base	DICE-RL	CAST
Can / drifting	87.7	94.0–97.0 [†]	97.8–99.3
Square / drifting	38.2	86.1–90.9	91.2–93.0
Square / diffusion	60.0	96.5	95.9
Square / flow matching	57.0	83.1	93.4^z
<i>External square references (separate protocols)</i>			
Method	Own base	Fine-tuned	Δ (pp)
DIPO [1]*	34.8	15.6	−19.2
QSM [2]*	34.8	0.0	−34.8
DQL *	34.8	0.0	−34.8
IDQL [3]*	34.8	32.7	−2.1
AWR [4]*	34.8	39.3	+ 4.5
DPPO [5]	60.0	67.8^o	+ 7.8

Table S1 retains all reported generator-class comparisons. On drifting square, CAST is ahead at the 20K-update comparison, while both actors approach 97–99% with longer training near 40K updates. The flow-matching square entry uses value-only guidance; its exact interaction budget is unspecified and no speed claim is drawn from it.

The diffusion square comparison uses a shared base with 60% success. The three first evaluated checkpoints at or above 90% success are 128K, 160K, and 160K steps for CAST, and 224K, 240K, and 256K for DICE-RL. Their means are $448/3 \approx 149.3\text{K}$ and 240K steps, a $1 - (448/3)/240 \approx 37.8\%$ reduction. These are discrete evaluation checkpoints, not exact crossing times interpolated from a learning curve. At 496K steps, CAST reaches 95.9% and DICE-RL reaches 96.5%. No confidence interval or continuous trajectory is inferred from these summary measurements.

Separate-protocol external references. DIPO, QSM, DQL, IDQL, and AWR use implementations adapted by the DPPO authors and approximately 24M environment steps. Their stricter rule requires success across an action chunk and scores the base at 34.8%. All gains in the external block use that same-harness base: for example, AWR’s 39.3%

is a 4.5 pp improvement. DPPO’s 67.8% comes from an ongoing run on a 60.0% base; a separate original-architecture run reaches 88.7% at 11.7M steps. Differences in bases, systems, budgets, and evaluation prevent interpreting these rows as an isolated test of CAST’s components.

S4 Component and Design Studies

Table S2: LIBERO success (%) with different restoration and clipping controls. Neither disables all controls; Clip enables only the two caps; Restore enables only radial restoration ($\eta_0 = 0$); CAST enables both restoring terms and both caps. Restore–CAST therefore changes weak restoration as well as clipping. Entries are task aggregates without training-variance estimates; Mean averages the five tasks. Bold marks the highest success per row, including the base.

Task	Base	Neither	Clip	Restore	CAST
Red mug	57.3	0.0	0.0	81.0	60.0
Ketchup	61.0	0.0	0.0	38.7	62.7
Bowl	81.8	0.0	0.0	80.7	84.0
Juice	78.5	0.0	0.0	86.0	82.0
Right caddy	57.8	0.0	0.0	42.3	63.0
Mean	67.3	0.0	0.0	65.7	70.3

Component settings and replication. The four configurations are specified in Table S3; Table S2 preserves their exact reported success rates. The source manuscript describes three seeds per cell, but the available aggregate record does not contain per-seed measurements or run manifests that verify training/evaluation seed linkage for every cell. We therefore show task aggregates without training-variance estimates. The main chart’s points are tasks, not seeds. Its means are computed from the five task values before rounding: base 67.28%, radial restoration 65.74%, and full CAST 70.34%. Their mean changes from the base are -1.54 and $+3.06$ pp, respectively.

Table S3: Controls enabled in the LIBERO component comparison. Caps limit the critic field and the combined displacement. Radial restoration only also disables weak restoration, so the study is not an exact two-factor ablation.

Configuration	Weak restoration	Radial restoration	Caps
No controls	Off	Off	Off
Clipping only	Off	Off	On
Radial restoration only	Off	On	Off
Full CAST	On	On	On

Alternative constraints. The loss-space hinge replaces the residual penalty’s shape with a penalty on excess RMS departure inside DICE-RL’s actor objective. It retains DICE-RL’s filtering and weighting. The LIBERO hard-projection recipe removes both restoring terms and both displacement caps, then projects the action target into a base-centered RMS ball. Thus, the alternatives change complete update rules rather than constraint location alone. The source reports three training seeds, with 300-episode square evaluations and 100 rollouts for each of three LIBERO evaluation seeds. The constraint campaign uses CAST references of 97/78/68% for square/mug/ketchup, distinct from the guidance-study references. The square value is the experimenter’s corrected measurement for this comparison; the other campaigns retain their reported values.

The LIBERO launchers identify the shared drifting multitask base, for which the source reports 57.3% on mug and 61.0% on ketchup. Both square design studies use the drifting base with 38.2% success. The constraint comparison reports 48.0% for the loss-space hinge and 97.0% for CAST. The guidance study retains its own CAST reference and auxiliary settings. The separate diffusion-base efficiency study still starts from 60.0% success.

Alternative guidance. The gradient recipe uses normalized critic action derivatives. Top- k transport uses the highest-valued current candidates as attractors; value-tilted transport assigns weights proportional to $\exp(\hat{q}_i/\tau)$, where $\hat{q}_i$ is standardized within the state’s candidate set. Both value-only recipes use kernel-weighted attraction and repulsion over sampled actions. In the available launchers, they also use a weak distributional pull toward independent base

samples, whereas the gradient recipe uses the paired pointwise weak pull. Historical square recipes additionally differ in radial-restoration settings. The reported square ranges span three training seeds. These comparisons do not isolate the source of critic guidance or establish that gradients always outperform rankings.

Restoration radius. Table S4 gives the exploratory single-training-seed sweep cited in the main text. The radius is measured in normalized RMS action units and controls when stronger restoration starts; it is not a hard bound on fitted policy outputs. The weak pull remains at infinite radius. These sweep references are retained separately from the main and component-study CAST results.

Table S4: LIBERO success (%) across restoration radii, with one training seed per setting. Radius ρ is measured in normalized RMS action units; $\rho = 0.05$ is this campaign’s reference. Infinite radius disables radial restoration while retaining the weak pull. These results do not establish an optimal radius.

Radius ρ (RMS)	Mug	Ketchup
0	54.9	60.1
0.02	86.3	73.4
0.05	78.1	68.4
0.10	81.2	69.4
0.20	77.1	67.1
∞	80.3	67.1

S5 Broader Evaluations

Table S5: DMC returns for refinements of a shared frozen base per task. Each evaluation averages ten 1000-step episodes (maximum return: 1000). The target budget is 400K environment steps; $\circ$ marks a run below budget, ν high evaluation variance, and t a tuned critic-guidance scale for the actor. Swing-up averages two training seeds; other tasks use one. Bold marks the highest reported return per row, including ties and runs below budget.

Task	Base	Unconstrained	DICE-RL	CAST
<i>Dense reward</i>				
Cartpole swing-up	429	575	318	715
Cartpole balance	378	815 $^\circ$	844	911
Quadruped run	238	97 $^\circ$	283	312^{ot}
Finger spin	3	0 $^\circ$	187°	187°
<i>Sparse reward</i>				
Cartpole balance	90	880 $^\circ$	843 $^\nu$	1000
Ball-in-cup catch	86	940°	66 $^\circ$	925

DMC evaluates one-step generative bases distilled from SAC experience. Table S5 retains dense/sparse groupings and all incomplete-run, tuning, and high-variance markers. Returns use the original 0–1000 scale. The target budget is 400K environment steps; exact budgets for the marked interim checkpoints are not available in the local records. Quadruped’s CAST return of 312 uses $\eta_Q = 1$ and is below budget; its default $\eta_Q = 0.5$ arm reports 267. The table is not a ranking of all methods at completed, matched budgets. In particular, the unconstrained sparse ball-in-cup interim return of 940 exceeds CAST’s 925. Sparse reward alone therefore does not imply failure without restoration.

Table S6: Success (%) in two comparisons where DICE-RL exceeds CAST. Robomimic transport uses matched budgets of 2.24M environment steps. Shared-critic LIBERO jointly refines over eight tasks and evaluates best-of-16 value-guided selection; this compares the complete systems under their respective refinement recipes.

Study	DICE-RL	CAST	Protocol
Robomimic transport	91.2	59.4	2.24M environment steps
Shared-critic LIBERO	72.2	69.2	Best-of-16; base 66.9%

Transport shows that restoration can accompany worse refinement even at a matched budget; it does not measure how far the optimal behavior lies from the base. Shared-critic LIBERO jointly trains over the eight tasks, unlike the main per-task study. The DICE-RL row uses its published recipe, so this is a system comparison. Single-candidate evaluation and best-of-16 selection give different conclusions: useful candidate ranking can remain even when residual learning changes little. Neither comparison establishes critic accuracy or a universal preference for stronger restoration.

References

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